

Steiner Routing

Implement the basic algorithm with the naïve search of the Steiner points. An example is provided in the supplement reading material. For GUI, show the routing trees for individual nets as well as the combined ones that show the overall routing graph usage.

A single input argument is needed for the filename. File format is:

1st line: <Grid size> <No. of nets>

All other lines: <net name> [<points> ...]

Each subsequent line is one net. The first field is the unique net name, followed by numbers in parenthesis () which are coordinates of the points (cells) in that net.

The first node in each net is the source node.

E.g., 10_30.nets: The grid size is 10x10, and there are 30 nets to be routed. The first point in each net is the source node for that net.

Your program should calculate the Manhattan wirelength for each net, and compute the total wirelength. Print this information on screen. Your program should also generate one output file showing the routing result. The output file format is:

1st line: <Grid size> <No. of nets>

All other lines: <net name> [<wire segments> ...]

The wire segment has four coordinates, (x1, y1), (x2, y2). The segment is from (x1,y1) to (x2,y2). E.g, net N1 which is from (1,1) through (2,2) to (3,2) should be written as:

N1 [(1,1),(2,2) (2,2),(3,2)]
